American Computer Science League

2025-2026 ● Contest 1: Short Problems Solutions ● Junior Division

1. Computer Number Systems

To convert from hexadecimal to octal, it is easiest to use 4 bits for hexadecimal and then 3 bits for octal.

$52326_{16} = 0101\ 0010\ 0011\ 0010\ 0110_2$
$= 01\ 010\ 010\ 001\ 100\ 100\ 110_2$
$= 1\quad 2\quad 2\quad 1\quad 4\quad 4\quad 6_8$
$= 1221446_8$

B. 1221446

2. Computer Number Systems

$(36_8 + 74_8) / 2$
$= 132_8 / 2$
$= (1*8^2 + 3*8 + 2) / 2$
$= (64 + 24 + 2) / 2$
$= 90_{10} / 2$
$= 45_{10}$

C. 45

3. Recursive Functions

$f(1) = 2$
$f(2) = 2 * f(2 - 1) + 1 = 2 * f(1) + 1 = 2 * 2 + 1 = 5$
$f(3) = 2 * f(3 - 1) + 1 = 2 * f(2) + 1 = 2 * 5 + 1 = 11$
$f(4) = 2 * f(4 - 1) + 1 = 2 * f(3) + 1 = 2 * 11 + 1 = 23$
$f(5) = 2 * f(5 - 1) + 1 = 2 * f(4) + 1 = 2 * 23 + 1 = 47$
$f(6) = 2 * f(6 - 1) + 1 = 2 * f(5) + 1 = 2 * 47 + 1 = 95$
Now substitute backwards.
The sequence is: 2, 5, 11, 23, 47, 95, …
Each term can be represented by $3 * 2^{n-1} - 1$.

D. 95

4. **Recursive Function**

$f(16) = f(16 - 3) + 16 = f(13) + 16 = 73 + 16 = 89$
$f(13) = f(13 - 3) + 13 = f(10) + 13 = 60 + 13 = 73$
$f(10) = 2 * f(10 - 2) = 2 * f(8) = 2 * 30 = 60$
$f(8) = 2 * f(8 - 2) = 2 * f(6) = 2 * 15 = 30$
$f(6) = 2 * 6 + 3 = 15$
Now substitute backwards.

E. 89

5. What Does This Program Do? (Branching)

```
input a, b
a = a ^ 2
b = b ^ 2
c = a + b
d = sqrt(c)
if d == int(d) then
    output a, b, c
else
    output "wrong"
end if
```

If a=5 and b=10, then $\sqrt{5^2 + 10^2} = \sqrt{125} \approx 11.18$. This isn't an integer. The others are all Pythagorean Triples: 3-4-5, 5-12-13, 9-12-15, and 7-24-25. The program prints a^2, b^2, and c^2. During this past year, 7/24/25 was Pythagorean Theorem Day.

C. a=5, b=10

6. What Does This Program Do? (Branching)

```
input num
if num % 3 == 0 &&
   num % 4 == 0 then
    output num
end if
if num % 5 == 0 ||
   num % 7 == 0 then
    output num
end if
```

For the numbers 1-25, inclusive, the following numbers are divisible by 3 and 4: 12, 24; the following numbers are divisible by either 5 or 7: 5, 10, 15, 20, 25, 7, 14, 21. There are 10 of them.

B. 10